

BTEC Level 3 Applied Science · Unit 1 Biology · Answer sheet**Cell Structure — Answers**

Mark scheme for the companion worksheet. Accept equivalent correct wording.

A

1. Nucleus: DNA / control of protein synthesis. Mitochondrion: aerobic respiration / ATP. Ribosome: protein synthesis.
2. Cell wall, chloroplasts, permanent vacuole.
3. Allows some substances to cross and restricts others.

B

4. Phospholipid bilayer with proteins (and cholesterol / glycoproteins in animal cells).
5. Folds increase surface area for respiratory enzymes / electron-transport proteins.
6. Support, shape, and prevention of osmotic bursting; freely permeable.

C

7. High ATP demand for contraction.
8. Photosynthesis occurs in green shoots and leaves; roots absorb water and minerals instead.
9. They have no nucleus, so no DNA for transcription.

D

10. Nucleus contains DNA and controls protein synthesis. Mitochondria carry out aerobic respiration / produce ATP.
11. Many chloroplasts; position near the upper leaf; vacuole maintains turgidity; elongated shape. Any two explained.
12. Wall: cellulose, freely permeable, support. Membrane: bilayer, selectively permeable, controls exchange.

E

13. RER synthesises and processes proteins. Golgi modifies and packages them into vesicles for secretion.